

Dynamic Guideline: WHO-Based Meningitis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO meningitis guideline

- **Guideline Title:** *WHO guidelines on meningitis diagnosis, treatment and care*
- **Publication Date:** 10 April 2025
- **Issuing Organization:** World Health Organization
- **Target Population:** Adults, adolescents, and children **older than 1 month** with **acute-onset, community-acquired meningitis**.
- **Scope and Objectives:** Evidence-based recommendations for diagnosis, antimicrobial treatment, adjunctive therapy, supportive care, post-exposure prophylaxis, rehabilitation, sequelae screening, and long-term care in community-acquired meningitis. WHO states that the guideline supports clinical management of children and adults with community-acquired meningitis, including acute and long-term care, and aligns with the *Defeating meningitis by 2030: a global road map* [WHO publication page](#).
- **Exclusions:** Neonatal meningitis, hospital-acquired meningitis, subacute or chronic meningitis, tuberculous meningitis, cryptococcal meningitis, and non-infectious meningitis are outside the guideline's stated scope [WHO EPI-WIN PDF](#).
- **Guideline Scale:** WHO describes this as its **first-ever comprehensive global clinical guideline on meningitis**, with **37 recommendations** based on **21 systematic reviews** [WHO EPI-WIN PDF](#).

Exactly 5 WHO meningitis-related guideline or guideline-level documents identified

#	WHO document	Date	Relevance to meningitis diagnosis, treatment, or care	Source
1	WHO guidelines on meningitis diagnosis, treatment and care	10 April 2025	Core WHO clinical guideline for community-acquired meningitis diagnosis, treatment, supportive care, prophylaxis, and sequelae management.	WHO publication page
2	Defeating meningitis by 2030: a global road map	Adopted by WHO Member States in 2020	Strategic framework for eliminating bacterial meningitis epidemics, reducing vaccine-preventable bacterial meningitis cases and deaths, and improving quality of life after meningitis.	ReliefWeb/WHO launch coverage
3	Fourth WHO model list of essential in vitro diagnostics	2023	Relevant to laboratory diagnosis capacity, including diagnostics needed for syndromes such as meningitis.	WHO IRIS guideline PDF
4	WHO model lists of essential medicines	2023	Relevant to access to essential antimicrobial and supportive medicines used in meningitis care.	WHO IRIS guideline PDF
5	Package of interventions for rehabilitation	2023	Relevant to rehabilitation and long-term sequelae management after meningitis.	WHO IRIS guideline PDF , APTA summary

Note on the requested “Meaningful youth engagement...” title

A WHO publication with the title pattern “**Meaningful youth engagement in a WHO guideline development process**” was identified, but it is **not a meningitis guideline**. The identified document is *Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)*, published **11 December 2025** by WHO Western Pacific [WHO Western Pacific publication page](#). The supplied evidence did **not** identify a WHO publication titled “*Meaningful youth*

engagement in a WHO guideline development process: case study of WHO meningitis guideline.” Its relevance to meningitis guidance is therefore methodological only, not disease-specific.

1.2 Key Recommendations

The table below prioritizes clinically actionable recommendations from WHO’s 2025 meningitis guideline. Where exact GRADE certainty was available from the summarized sources, it is reported directly. Where a WHO recommendation was visible but the exact certainty was not available in the supplied excerpts, this is marked transparently.

Clinical Area	Recommendation	GRADE Evidence Quality	Strength
Lumbar puncture	In suspected acute meningitis, perform lumbar puncture as soon as possible, preferably before antimicrobial treatment, unless there are contraindications or reasons for deferral.	Not applicable to good practice statement	Good practice statement
CSF Gram stain	In suspected acute meningitis, perform Gram stain on cerebrospinal fluid.	Moderate	Strong
CSF routine biochemistry and cytology	In suspected acute meningitis, perform CSF white blood cell count with differential, protein, glucose, and CSF-to-blood glucose ratio.	Moderate	Strong
CSF lactate	Consider CSF lactate when antibiotic therapy has not yet been administered.	Moderate	Conditional
Cranial imaging	Do not perform cranial imaging routinely in suspected acute meningitis.	Very Low	Strong against

Clinical Area	Recommendation	GRADE Evidence Quality	Strength
Cranial imaging before LP when readily accessible	If cranial imaging is readily accessible, perform imaging before LP when high-risk features are present: Glasgow Coma Score below 10, focal neurological signs, cranial nerve deficits, papilloedema, new-onset seizures in adults, or severe immunocompromise.	Very Low	Strong
LP deferral when imaging unavailable	If cranial imaging is not readily accessible, defer LP when the same high-risk features are present until they resolve.	Very Low	Strong
Treatment timing	Do not delay treatment for cranial imaging or when LP is deferred.	Very Low	Strong
Empiric antibiotics	Give IV ceftriaxone or cefotaxime as empiric treatment in children and adults with suspected or probable acute bacterial meningitis. Add ampicillin or amoxicillin if risk factors for <i>Listeria monocytogenes</i> are present.	Exact certainty not reported in supplied excerpts	Strong in guideline summary
Adjunctive corticosteroids	In non-epidemic settings where LP can be performed, initiate IV corticosteroids with the first antibiotic dose in suspected acute bacterial meningitis; for suspected or probable pneumococcal meningitis, dexamethasone is the example cited.	Exact certainty not reported in supplied excerpts	Strong in WHO webinar summary
Raised intracranial pressure	Use osmotic agents such as mannitol or hypertonic saline temporarily for increased intracranial pressure in bacterial meningitis, especially with signs of impending herniation.	Low	Conditional
Seizures	In acute symptomatic seizures due to meningitis, continue antiseizure medicines	Exact certainty not reported in	Conditional

Clinical Area	Recommendation	GRADE Evidence Quality	Strength
	for no longer than 3 months if seizures do not recur.	supplied excerpts	
Meningococcal prophylaxis, sporadic disease	Provide single-dose parenteral ceftriaxone or oral ciprofloxacin to close contacts of laboratory-confirmed meningococcal disease, guided by antimicrobial susceptibility.	Very Low	Strong
Meningococcal prophylaxis, large-scale epidemic	During large-scale epidemics, provide single-dose ceftriaxone or ciprofloxacin to close contacts of clinically suspected meningococcal disease, guided by antimicrobial susceptibility.	Very Low	Strong
Sequelae assessment	Review children and adults with acute meningitis for sequelae before discharge and at follow-up.	Exact certainty not reported in supplied excerpts	Strong
Audiological screening	Conduct formal audiological screening before discharge or, if not possible, within 4 weeks after discharge.	Exact certainty not reported in supplied excerpts	Strong
Rehabilitation	Provide rehabilitation as soon as possible for children and adults with sequelae due to acute meningitis from any cause.	Exact certainty not reported in supplied excerpts	Strong
Hearing rehabilitation	Provide early hearing rehabilitation for people with hearing loss after meningitis.	Exact certainty not reported in supplied excerpts	Strong

Sources: WHO publication page and annex description [WHO publication page](#); WHO EPI-WIN overview [WHO EPI-WIN PDF](#); NCBI Bookshelf recommendation excerpts [NCBI Bookshelf recommendations](#); WHO guideline PDF/IRIS [WHO IRIS PDF](#); Medscape summary [Medscape summary](#).

1.3 Guideline Development Methodology

Evidence Review Process

WHO developed the 2025 meningitis guideline according to WHO guideline development standards and the WHO handbook methodology [NCBI methodology chapter](#).

Key methodological features:

- **20 PICO-format guideline questions** covering diagnosis, treatment, supportive care, prophylaxis, and long-term care.
- **21 systematic reviews**, including **20 quantitative evidence reviews** and **1 qualitative evidence review** [WHO EPI-WIN PDF](#).
- Use of the **GRADE approach** and **Evidence-to-Decision frameworks**.
- **Web Annex A:** quantitative evidence reports.
- **Web Annex B:** qualitative and economic evidence.
- **Web Annex C:** Evidence-to-Decision frameworks [WHO publication page](#).
- For randomized trials, WHO review teams used the **Cochrane Risk of Bias 2 tool** [WHO IRIS PDF](#).

Consensus Method

The recommendation-development process involved:

- WHO Steering Group
- Guideline Development Group
- Guideline methodologist
- WHO Secretariat
- Evidence review teams
- External Review Group

The available sources confirm that recommendations were developed through WHO's structured guideline process using GRADE Evidence-to-Decision frameworks, but they do not specify whether final decisions were made by formal vote, unanimity, or another named consensus rule [WHO EPI-WIN PDF](#), [NCBI methodology chapter](#).

Conflict of Interest Management

Conflicts of interest were handled according to WHO procedures:

- Contributors submitted declarations of interest.
- WHO reviewed declarations, curricula vitae, and biographies.
- Declared interests were assessed and managed.
- Declarations were updated throughout the process.
- A summary of declared interests for Guideline Development Group and External Review Group members was included in guideline annex materials [NCBI methodology chapter](#).

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on meningitis diagnosis, treatment and care: executive summary) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was synthesized for the period after release of the WHO meningitis guideline on **10 April 2025** through **29 April 2026**, focusing on:

- New clinical guidelines or major guideline updates.
- Systematic reviews and meta-analyses.
- Randomized or interventional trials.
- Major surveillance studies.
- Authoritative institutional updates from WHO, CDC, NICE, IDSA, and clinical trial registries.
- Cryptococcal and fungal meningitis evidence only where relevant, while noting that WHO 2025 acute meningitis guidance excludes cryptococcal meningitis.

Databases and Sources Searched

- WHO publications and WHO news pages.
- WHO EPI-WIN materials.
- NCBI Bookshelf.
- NICE guideline portal.
- CDC clinical and Emerging Infectious Diseases pages.

- [ClinicalTrials.gov](https://clinicaltrials.gov).
- IDSA guideline portal.
- BMJ and other peer-reviewed medical summaries identified in the source set.

Inclusion Criteria

Included:

- Evidence or guidance published, updated, or newly identified after **10 April 2025** and before or on **29 April 2026**.
- Evidence relevant to meningitis recognition, diagnosis, antimicrobial treatment, adjunctive therapy, supportive care, prophylaxis, sequelae, rehabilitation, or surveillance.
- High-authority institutional guidance and trial registry records.

Excluded:

- Non-meningitis evidence.
- Opinion-only sources without clinical, epidemiologic, or guideline relevance.
- Neonatal, tuberculous, cryptococcal, fungal, or hospital-acquired meningitis evidence when presented as if directly covered by the WHO 2025 acute community-acquired meningitis guideline; such evidence is discussed separately where relevant.

2.2 Key New Studies

Study or Source	Design	Key Findings	GRADE Quality	Recommendation Strength
CDC <i>Emerging Infectious Diseases</i> : Group A Streptococcus meningitis in the United States, 1997–2022; updated 26 January 2026	Long-term observational surveillance analysis	Evaluated 26 years of Active Bacterial Core surveillance data. Authors concluded that Group A Streptococcus should be included in future meningitis studies, reviews, and updated clinical	Low	No direct recommendation; evidence supports surveillance and future guideline consideration

Study or Source	Design	Key Findings	GRADE Quality	Recommendation Strength
		practice guidelines.		
ClinicalTrials.gov NCT06666322: Platform Trial for Cryptococcal Meningitis; updated 7 January 2026	Phase 2/3 interventional platform trial registry record	Trial in HIV- associated cryptococcal meningitis; estimated enrollment 2,000; interventions include standard care, oteseconazole, Sfu-AM2-19 injection, and additional antifungal arms. No results posted.	Not assessable; no results available	No practice recommendation
NICE NG240 PDF marked © NICE 2026; original review 19 March 2024	Guideline update/maintenance document	NICE continues to cover bacterial meningitis and meningococcal disease recognition, diagnosis, and management in babies, children, young people, and adults; excludes confirmed fungal, viral, tuberculous meningitis and immunodeficiency populations.	Moderate for guideline consistency; not new trial evidence	Supports rapid recognition and treatment; no change to WHO recommendations

Study or Source	Design	Key Findings	GRADE Quality	Recommendation Strength
CDC meningococcal disease clinical overview	Public health clinical guidance	CDC emphasizes prompt diagnosis and treatment, chemoprophylaxis for close contacts regardless of vaccination status, and vaccination for all preteens/teens and higher-risk groups.	Moderate for public health consistency; not a new comparative trial	Strong public health practice alignment
Global cryptococcosis guideline / cryptococcal meningitis management evidence	International fungal disease guideline	Recommends careful exclusion of relapse and other diagnoses in suspected cryptococcal IRIS; MRI and LP with opening pressure; therapeutic LP; continuation of antifungal therapy; selected corticosteroids for persistent symptoms.	Low to Moderate, depending on question; outside WHO 2025 meningitis scope	Conditional to strong depending on cryptococcal-specific context
WHO “Meaningful youth engagement...” evidence brief, 11 December 2025	Methodological case study, not meningitis-specific	Describes youth engagement in development of WHO abortion care guideline; no direct meningitis clinical recommendations.	Very Low for meningitis applicability	No meningitis recommendation

Sources: CDC clinical overview [CDC clinical overview](#); CDC EID article [CDC EID article](#); ClinicalTrials.gov [ClinicalTrials.gov NCT06666322](#); NICE overview [NICE overview](#); NICE PDF [NICE PDF](#); cryptococcosis guideline [Global cryptococcosis guideline](#); WHO youth engagement case study [WHO Western Pacific publication page](#).

2.3 Evidence Gaps and Future Research Needs

1. Full post-guideline comparative treatment evidence remains limited

No completed randomized trial after **10 April 2025** with practice-changing results for acute community-acquired bacterial meningitis was identified in the supplied evidence. Future priorities include:

- Optimal empiric regimens in regions with changing antimicrobial resistance.
- Adjunctive corticosteroid benefit by pathogen, HIV status, age, and timing.
- Best supportive care protocols for raised intracranial pressure in low-resource settings.
- Seizure management duration after acute symptomatic seizures.

2. Cryptococcal meningitis remains outside the WHO 2025 acute meningitis scope

WHO 2025 excludes cryptococcal meningitis, but cryptococcal meningitis remains a major cause of meningitis-related mortality in people with advanced HIV. The Phase 2/3 platform trial [NCT06666322](#) is important, but no results were available by **29 April 2026** [ClinicalTrials.gov NCT06666322](#).

3. Surveillance evidence suggests pathogen evolution

The CDC EID analysis suggests that Group A Streptococcus meningitis deserves more attention in future studies and guideline updates [CDC EID article](#). This does not immediately change empiric therapy but supports pathogen-inclusive surveillance and future guideline review.

4. Long-term sequelae care needs stronger implementation evidence

WHO strongly recommends sequelae assessment, audiological screening, rehabilitation, and hearing rehabilitation, but implementation studies are needed, especially in low- and middle-income countries where access to hearing services, rehabilitation, educational support, and disability services may be limited. WHO notes that meningitis survivors may experience deafness, learning impairment, behavioural problems, and psychosocial consequences, and that services are often insufficient in low- and middle-income countries [WHO meningitis fact sheet](#).

5. Youth and community engagement methods are not yet meningitis-specific

WHO's youth engagement case study is methodologically relevant but not meningitis-specific [WHO Western Pacific publication page](#). Future meningitis guideline updates could consider structured involvement of young survivors, caregivers, disability advocates, and communities affected by meningitis epidemics.

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact Through 29 April 2026	Updated Recommendation
Perform LP as soon as possible in suspected acute meningitis unless contraindicated or deferred.	No new evidence identified that changes this recommendation.	Maintain WHO recommendation. Do not delay antimicrobial therapy when LP is unsafe or delayed.
Perform CSF Gram stain, WBC count, protein, glucose, and CSF-to-blood glucose ratio.	No new contradictory evidence identified.	Maintain WHO recommendation. Strengthen laboratory capacity through essential diagnostics planning.
Do not perform cranial imaging routinely; image before LP only when indicated by high-risk features where imaging is accessible.	No new evidence identified that changes this risk-based approach.	Maintain WHO recommendation. Treatment must not wait for imaging.
Start empiric IV ceftriaxone or cefotaxime for suspected or probable acute bacterial meningitis; add ampicillin/amoxicillin when <i>Listeria</i> risk factors exist.	CDC Group A Streptococcus surveillance highlights evolving pathogen considerations but does not require a change to standard empiric therapy based on available data.	Maintain WHO empiric therapy. Countries should adapt empiric regimens to local pathogen distribution and resistance data.
Use adjunctive corticosteroids in selected bacterial meningitis contexts, especially	No new post-guideline trial results identified.	Maintain WHO recommendation. Benefit depends on early administration and likely pathogen.

Original WHO Recommendation	New Evidence Impact Through 29 April 2026	Updated Recommendation
suspected/probable pneumococcal disease, with the first antibiotic dose.		
Use osmotic agents such as mannitol or hypertonic saline temporarily for raised intracranial pressure.	No new comparative evidence identified.	Maintain conditional recommendation. Use as temporizing therapy while arranging definitive critical care support when available.
Provide meningococcal chemoprophylaxis to close contacts according to epidemiologic context.	CDC guidance aligns with close-contact prophylaxis regardless of vaccination status CDC clinical overview .	Maintain WHO recommendation. Ensure prophylaxis is rapid and guided by susceptibility patterns.
Assess for sequelae before discharge and at follow-up; provide audiological screening and rehabilitation.	No new evidence changes the recommendation; WHO burden data continue to support strong implementation.	Maintain and operationalize as a discharge standard: hearing screen, neurologic/developmental assessment, rehabilitation referral, and follow-up plan.
Cryptococcal meningitis is outside WHO 2025 acute community-acquired meningitis scope.	Phase 2/3 cryptococcal meningitis platform trial is ongoing with no results by cutoff.	No change to WHO meningitis guideline. Use cryptococcal-specific guidance for suspected cryptococcal meningitis.

Practical Clinical Implementation Summary

1. Treat suspected bacterial meningitis as an emergency.

- Do not delay antibiotics for imaging or lumbar puncture when LP is unsafe or delayed.

2. Prioritize early CSF diagnosis when safe.

- LP should be done as soon as possible unless contraindications or deferral criteria are present.

3. Use empiric IV ceftriaxone or cefotaxime.

- Add ampicillin or amoxicillin when *Listeria monocytogenes* risk factors are present.

4. Use cranial imaging selectively.

- Routine imaging is not recommended.
- Image before LP only when indicated by depressed consciousness, focal signs, papilloedema, new-onset adult seizures, cranial nerve deficits, or severe immunocompromise.

5. Give adjunctive corticosteroids early when indicated.

- Start with the first antibiotic dose in appropriate suspected bacterial meningitis contexts, especially suspected pneumococcal disease.

6. Plan discharge around sequelae prevention and detection.

- Hearing screening before discharge or within 4 weeks.
- Neurologic, cognitive, developmental, behavioural, and functional sequelae review.
- Early rehabilitation and hearing rehabilitation when needed.

7. Use meningococcal prophylaxis for close contacts.

- Ceftriaxone or ciprofloxacin, guided by setting and susceptibility patterns.

8. Do not apply WHO 2025 acute meningitis guidance to cryptococcal meningitis.

- Use cryptococcal-specific guidelines and monitor ongoing trial evidence.